

Townsend 3.1-3.5

Questions for Gallicchio

I am very confused by the eigenstates and operators for angular momentum when shown in matrix form, even though it looks very clean in $\langle | \rangle$ notation.

What are the uses/physical correspondence to the raising and lowering operators?

What is j for the states as basis - dimension? Something weird?

Rotators Do Not Commute, and Neither Do Generators

Rotations don't commute. Generators make rotations, so they don't either.

Commutator of two operators:

$$\begin{aligned} [\hat{J}_x, \hat{J}_y] &= i\hbar \hat{J}_z \\ [\hat{J}_z, \hat{J}_x] &= i\hbar \hat{J}_y \end{aligned}$$

These tell us about eigenstates of angular momentum operators.

3.2 Commuting Operators

If rotations commute, then

$$[\hat{A}, \hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A} = 0$$

And they both have eigenvalues corresponding to an eigenstate

If $\hat{A}|a\rangle = a|a\rangle$ then $\hat{B}|a\rangle = b|a\rangle$

so we relabel the state

$|a, b\rangle$

3.3 Eigenvalues and Eigenstates of Angular momentum

Angular momentum operators:

$$\hat{J}_x \rightarrow \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \hat{J}_y \rightarrow \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad \hat{J}_z \rightarrow \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad (3.28)$$

Raising and lowering operators have eigenvalues of $m + 1$ or $m - 1$.

They just step up/down angular momentum by 1 unit of $\hbar$.

3.4 Matrix elements of raising and lowering operators

We have a thing called j (dimension?) which make all the basis states, and from those we generate the matrix representations of raising and lowering operators.

3.5 Uncertainty Relations and Angular Momentum

For non commutative operators we have

$$\Delta J_x \Delta J_y \geq \frac{\hbar}{2} |\langle J_z \rangle|$$